IT4100

Software Quality Assurance

Group Assignment 1

Member Details

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SQA Group ID	SQA-2026-170
Module	Image Converter module
Automation Tool	TestNG

1. Objective

The objective of this section is to test the **Image Converter** functionality of the Pixelssuite web application using **manual testing** and **TestNG automation**.

The testing focuses on validating key user interactions, including page accessibility, initial user interface visibility, valid image upload, image conversion from JPG to PNG, and download functionality.

Additionally, the goal is to ensure that the user interface responds correctly and that the core features operate reliably under normal usage conditions.

2. Test Environment

- **Browser:** Google Chrome
- **Target URL:** https://www.pixelssuite.com/
- **Selected Feature:** Image Converter → To PNG
- **Testing Type:** Functional Testing, Smoke Testing, Basic Negative Testing
- **Automation Tool:** Selenium WebDriver with TestNG
- **Programming Language:** Java
- **Development Environment:** Visual Studio Code
- **Operating System:** Windows

3. Manual Test Cases for the Image Converter Module.

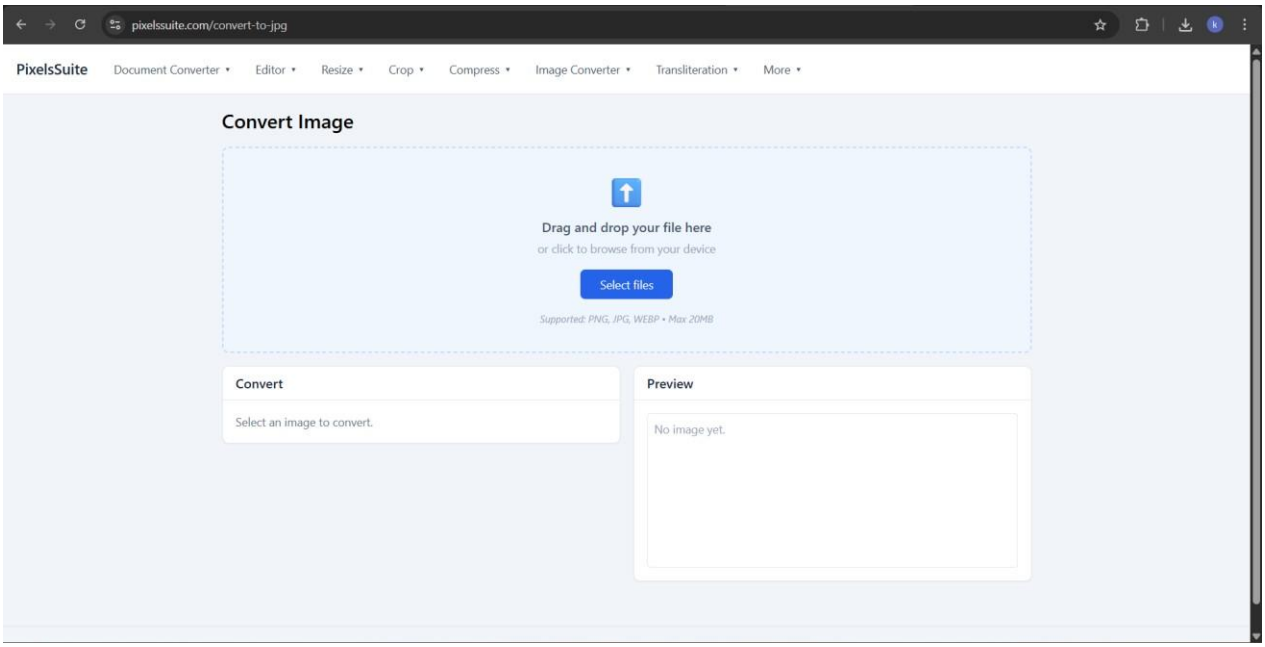
Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
M/T-01	Access Image Converter page	1. Open the Pixelssuite website. 2. Navigate to Image Converter → To PNG.	The Image Converter page loads successfully.	The Image Converter page loaded successfully and the main sections were displayed properly.	Pass
M/T-02	Verify initial UI state	1. Observe the upload area, convert section, and preview section before	Upload panels are visible. Convert section shows default message.	Initial UI matched the expected empty state.	Pass

		uploading any file.	Preview area is empty.		
M/T-03	Upload valid JPG image	1. Click Select files. 2. Choose a valid JPG image from the local computer.	The selected JPG image is uploaded successfully and preview is displayed.	The selected JPG image was uploaded successfully. The image preview was displayed, and the Convert section became active.	Pass
M/T-04	Convert JPG to PNG	1. Upload a valid JPG image. 2. Start the conversion process.	The system converts the JPG image to PNG successfully.	The system processed the uploaded JPG image correctly for PNG conversion. The format was shown as **JPG to PNG** and the preview was displayed successfully.	Pass
M/T-05	Download converted PNG image	1. Complete the image conversion. 2. Click Download.	The converted PNG image is downloaded successfully.	The converted PNG image was downloaded successfully. The browser download notification showed **image.png** with status **Done**.	Pass
M/T-06	Reset using Clear button	1. Open Image Converter → To PNG 2. Upload a valid JPG image 3. Click **Clear**	The page should return to the initial empty state and remove the uploaded image from both the Convert and Preview sections.	After clicking **Clear**, the Convert section was reset, but the Preview section still displayed the previously uploaded image.	Fail

4. Screenshot Evidence

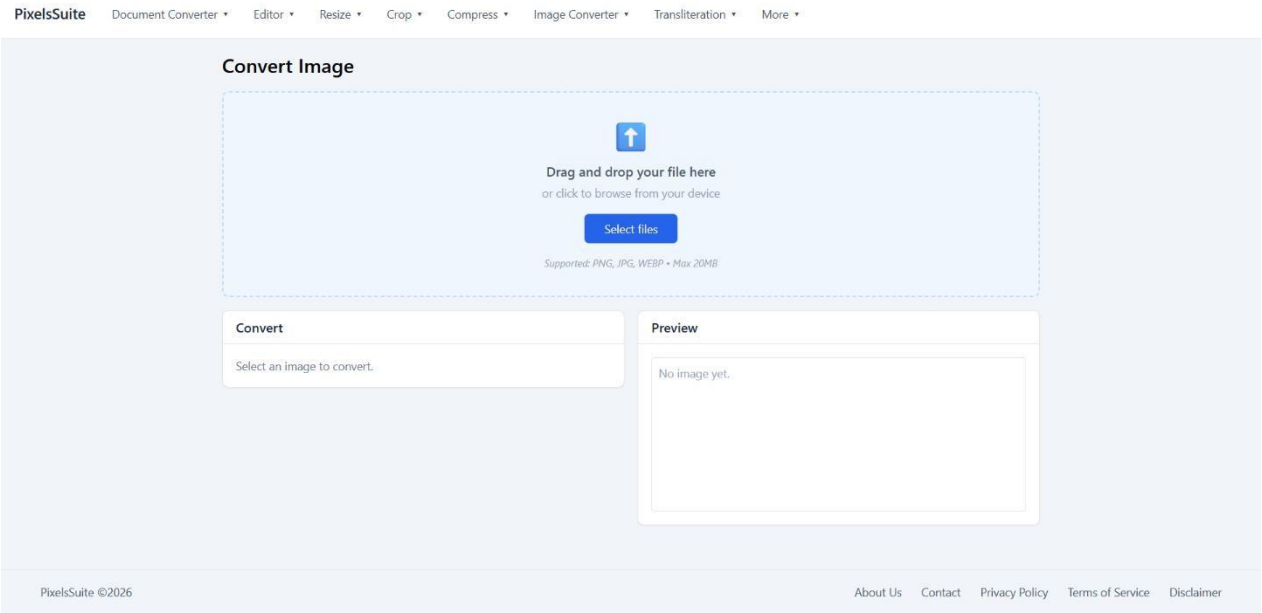
Test Case ID: - M/T-01

Test Scenario: - Access Image Converter page



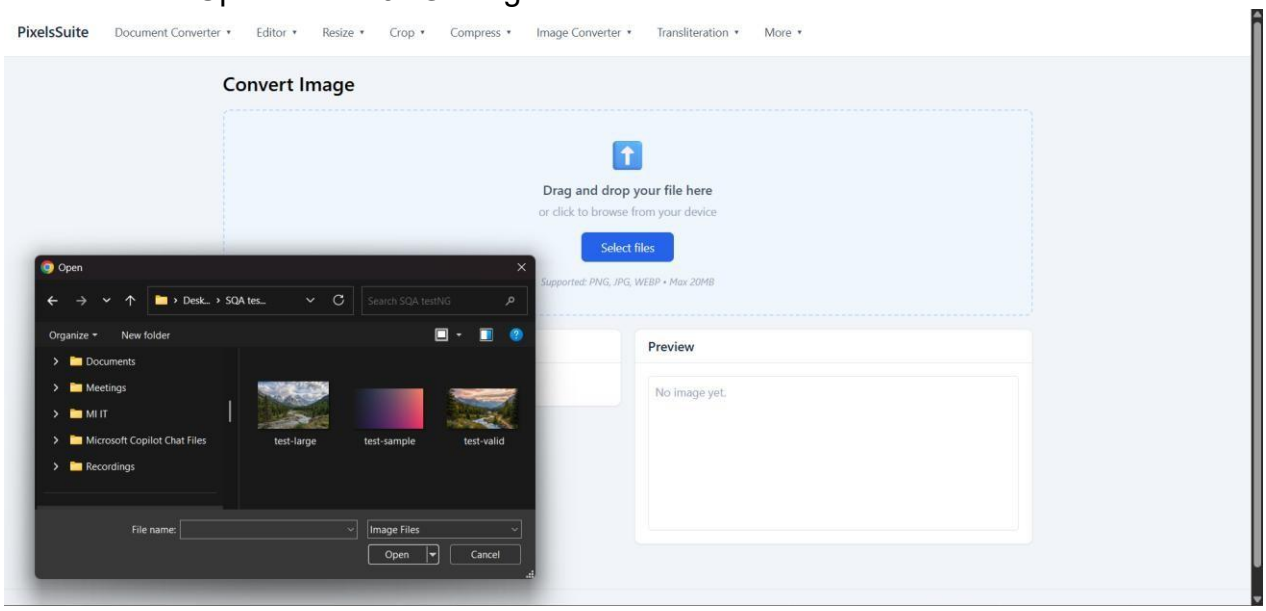
Test Case ID: - M/T-02

Test Scenario: - Verify initial UI state



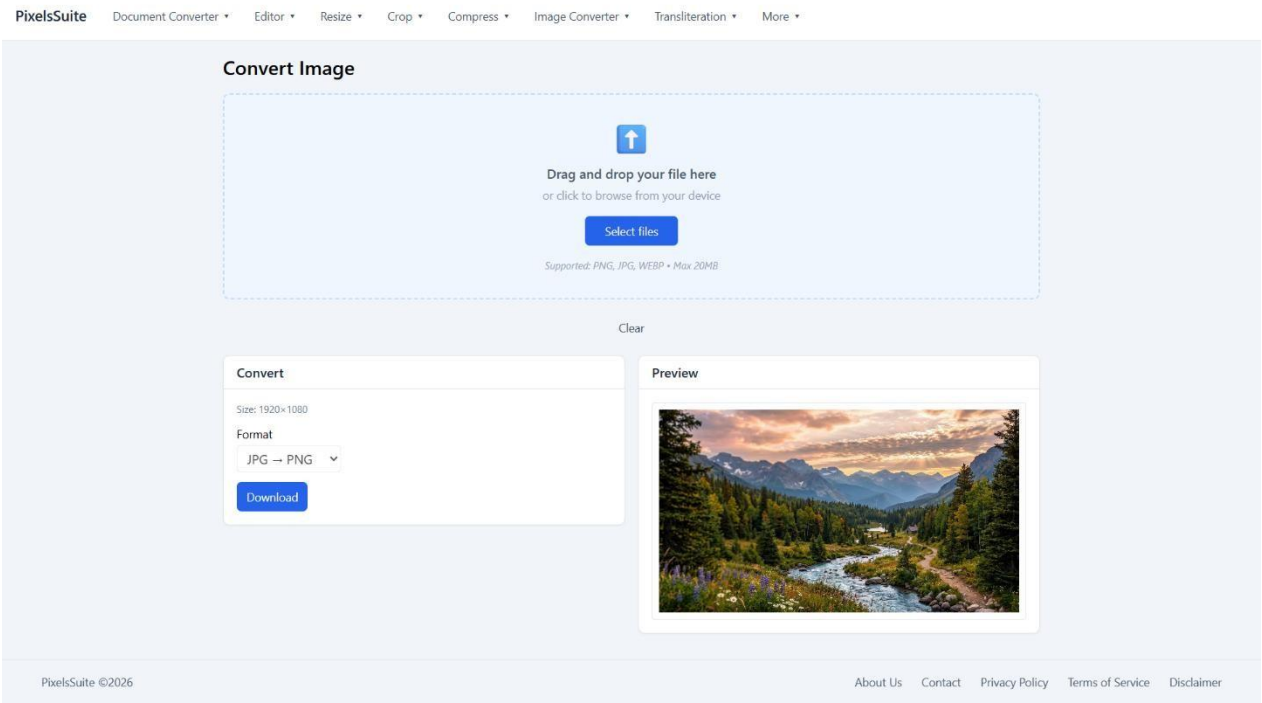
Test Case ID: - M/T-03

Test Scenario: - Upload valid JPG image



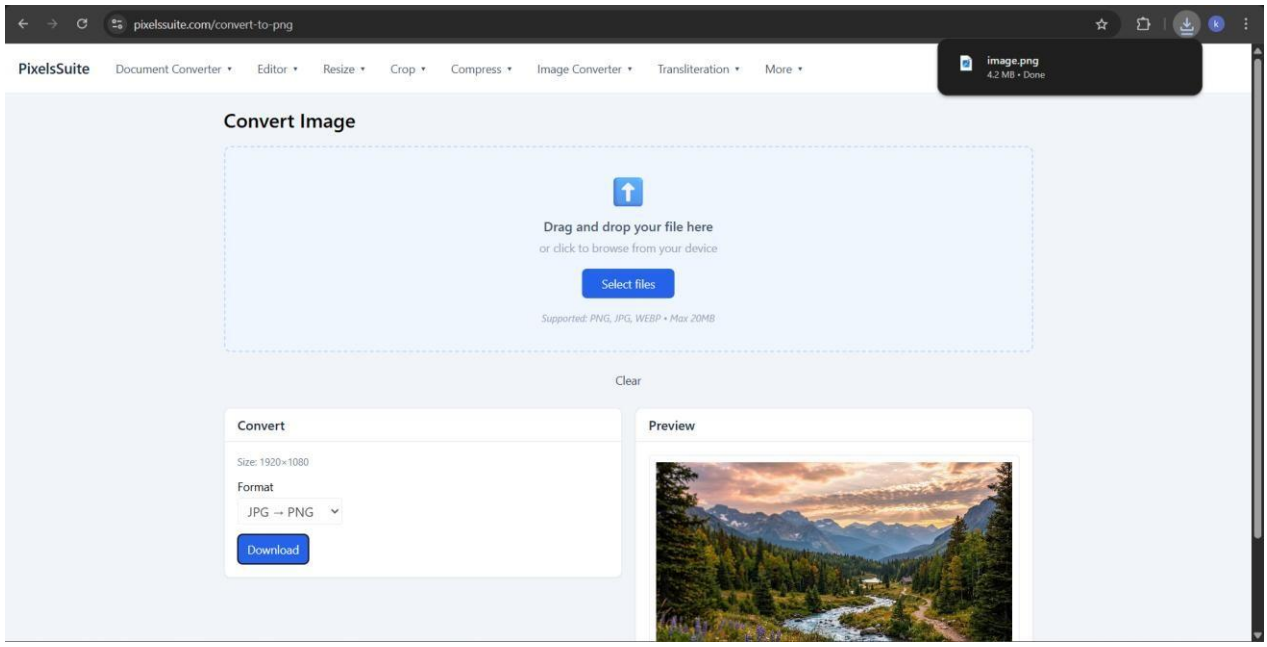
Test Case ID: - M/T-04

Test Scenario: - Convert JPG to PNG



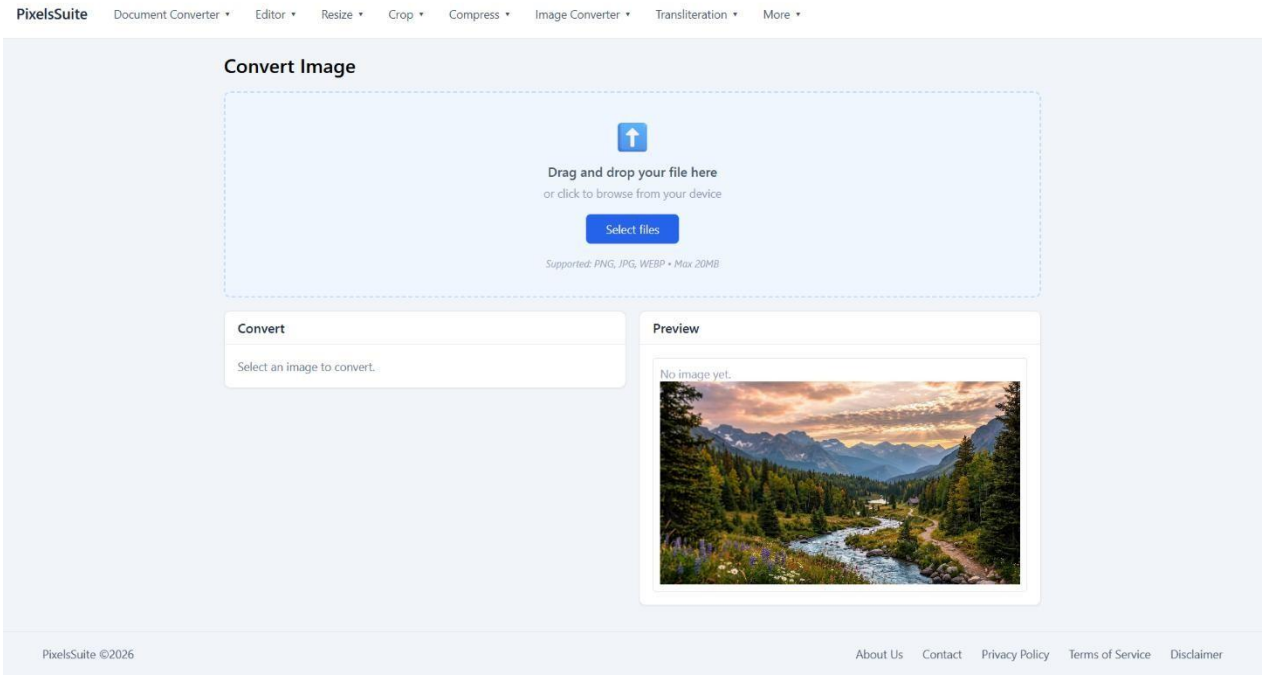
Test Case ID: - M/T-05

Test Scenario: - Download converted PNG image



Test Case ID: - M/T-06

Test Scenario: - Reset using Clear button



5. Automated Testing – TestNG

5.1 Code Implementation

The automated test suite was developed using **Java, Selenium WebDriver, and TestNG** within **Visual Studio Code**.
The automation script was created for the **Image Converter** module of the Pixelssuite web application.
The script covers the main functional scenarios of the selected feature, including:

- page loading
- valid JPG image upload

Reusable setup and teardown methods were used for browser initialization and closing.
Assertions were implemented using **TestNG** to validate whether the page loaded successfully and whether the upload process worked correctly.

crop_jpg_playwright.spec.js - playwright - Visual Studio Code

PLAYWRIGHT

- node_modules
- playwright-report
- test-results
- tests
  - fixtures
  - crop_jpg_playwright.spe...
- package-lock.json
- package.json
- playwright.config.js

crop_jpg_playwright.spec.js

tests > crop_jpg_playwright.spec.js > cropImage

```
91  }
92
93  test('TC_CROP_001 Access Crop JPG page', async ({ page }) => {
94    await openCropPage(page);
95
96    // Screenshot point: successful page entry.
97    await expect(cropHeading(page)).toBeVisible({ timeout: 15000 });
98    await expect(selectFilesButton(page)).toBeVisible();
99    await expect(hiddenFileInput(page).first()).toBeAttached();
100
101   // Confirms the expected route loaded.
102   await expect(page).toHaveURL(/\/crop-jpg$/i);
103 });
104
105 test('TC_CROP_002 Verify initial UI state', async ({ page }) => {
106   await openCropPage(page);
107
108   // Screenshot point: initial/default state before upload.
109   await expect(selectFilesButton(page)).toBeVisible();
110   await expect(originalText(page)).not.toBeVisible();
111
112   // Empty crop/default state checks without relying only on one exact placeholder text.
113   expect(await cropValueInputs(page).count()).toBe(0);
114   await expect(cropImage(page)).not.toBeVisible();
115
116   // Preview should still be in default state before upload.
117   // Some builds keep a canvas node mounted, so we avoid asserting strict DOM absence.
118   await expect(downloadButton(page)).not.toBeVisible();
119   await expect(clearText(page)).not.toBeVisible();
120 });
121
122 test('TC_CROP_003 Open file selector', async ({ page }) => {
123   await openCropPage(page);
124
125   // Intended to open the operating system file picker dialog.
126   // The visible "Select files" label is nested; click its wrapper for reliability.
127   await selectFilesButton(page).first().locator('xpath=..').click({ force: true });
128
129   // Minimal stable assertion after click path.
130   await expect(hiddenFileInput(page).first()).toBeAttached();
131
      Implement with Codex
132   // TODO: native OS file dialogs are outside Playwright DOM assertions.
133   // Upload itself is validated with setInputFiles in later test cases.
134 });
135
136 test('TC_CROP_004 Upload valid JPG image', async ({ page }) => {
137   await openCropPage(page);
138   await uploadFixtureImage(page);
```

Screen Reader Optimized | Ln 46, Col 2 | Spaces: 2 | UTF-8 | LF | JavaScript | Port : 5500 | Prettier

crop_jpg_playwright.spec.js - playwright - Visual Studio Code

PLAYWRIGHT

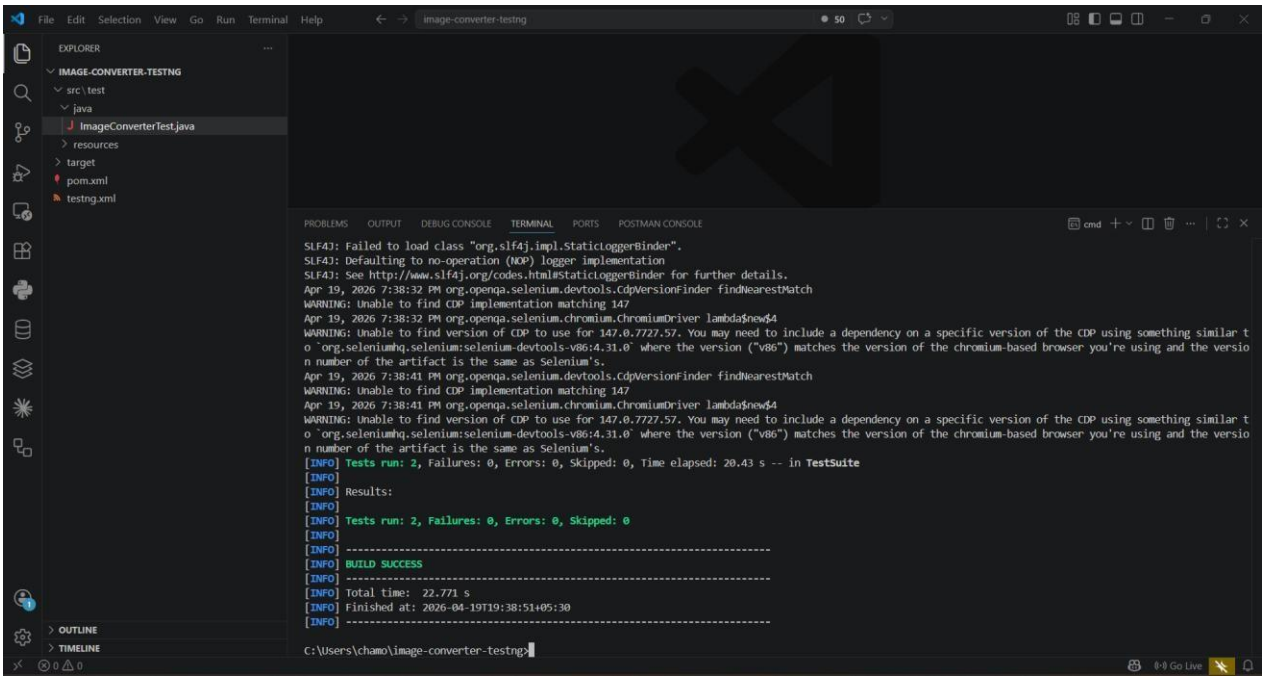
- node_modules
- playwright-report
- test-results
- tests
  - fixtures
  - crop_jpg_playwright.spe...
- package-lock.json
- package.json
- playwright.config.js

crop_jpg_playwright.spec.js

tests > crop_jpg_playwright.spec.js > cropImage

```
122 test('TC_CROP_003 Open file selector', async ({ page }) => {
133   // Upload itself is validated with setInputFiles in later test cases.
134 });
135
136 test('TC_CROP_004 Upload valid JPG image', async ({ page }) => {
137   await openCropPage(page);
138   await uploadFixtureImage(page);
139
140   // Screenshot point: upload result in crop + preview panes.
141   await expect(originalText(page)).toBeVisible();
142   await expect(cropImage(page)).toBeVisible();
143   await expect(cropSelectionBox(page)).toBeVisible();
144   await expect(previewCanvas(page)).toBeVisible();
145   await expect(clearText(page)).toBeVisible();
146 });
147
148 test('TC_CROP_005 Verify crop value text boxes after upload', async ({ page }) => {
149   await openCropPage(page);
150   await uploadFixtureImage(page);
151
152   const { xInput, yInput, widthInput, heightInput } = await getFirstFourVisibleCropInputs(page);
153
154   // Screenshot point: the first four non-file inputs interpreted as X/Y/Width/Height.
155   await expect(xInput).toBeVisible();
156   await expect(yInput).toBeVisible();
157   await expect(widthInput).toBeVisible();
158   await expect(heightInput).toBeVisible();
159
160   await expect(xInput).toHaveValue(/.+/);
161   await expect(yInput).toHaveValue(/.+/);
162   await expect(widthInput).toHaveValue(/.+/);
163   await expect(heightInput).toHaveValue(/.+/);
164 });
165
166 test('TC_CROP_006 Edit crop using text boxes', async ({ page }) => {
167   await openCropPage(page);
168   await uploadFixtureImage(page);
169
170   const { xInput, yInput, widthInput, heightInput } = await getFirstFourVisibleCropInputs(page);
171
172   // Screenshot point: manual crop values edited via text boxes.
173   await xInput.fill('5');
174   await expect(xInput).toHaveValue('5');
175
176   await yInput.fill('5');
177   await expect(yInput).toHaveValue('5');
178
179   await widthInput.fill('70');
180   await expect(widthInput).toHaveValue('70');
```

Screen Reader Optimized | Ln 46, Col 2 | Spaces: 2 | UTF-8 | LF | JavaScript | Port : 5500 | Prettier


File Edit Selection View Go Run Terminal Help
image-converter-testng
EXPLORER
IMAGE-CONVERTER-TESTNG
src\test
java
ImageConverterTest.java
resources
target
pom.xml
testng.xml
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Apr 19, 2026 7:38:32 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find CDP implementation matching 147
Apr 19, 2026 7:38:32 PM org.openqa.selenium.chromium.ChromiumDriver lambdanew4
WARNING: Unable to find version of CDP to use for 147.0.7727.57. You may need to include a dependency on a specific version of the CDP using something similar to `org.seleniumhq.selenium:selenium-devtools-v86:4.31.0` where the version ("v86") matches the version of the chromium-based browser you're using and the version number of the artifact is the same as Selenium's.
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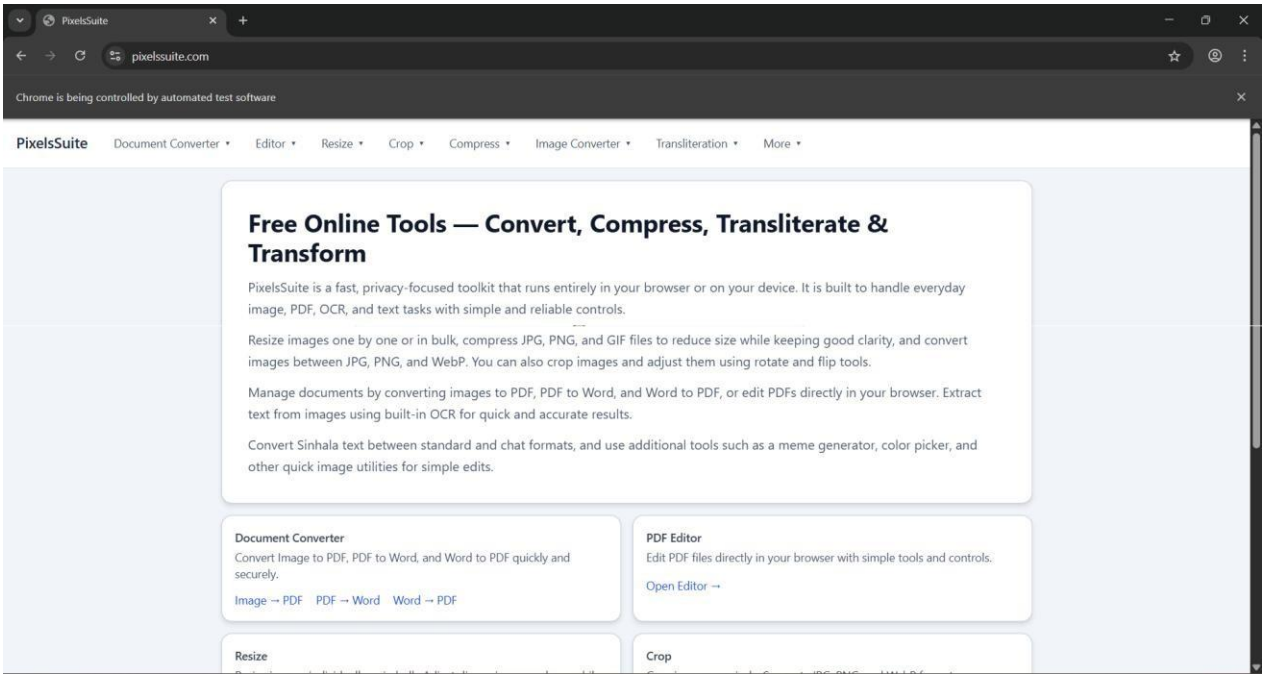
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 20.43 s -- in TestSuite
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --
[INFO] BUILD SUCCESS
[INFO] --
[INFO] Total time: 22.771 s
[INFO] Finished at: 2026-04-19T19:38:51+05:30
[INFO] --
C:\Users\chamo\image-converter-testng>


PixelsSuite
pixelssuite.com
Chrome is being controlled by automated test software
PixelsSuite
Document Converter
Editor
Resize
Crop
Compress
Image Converter
Transliteration
More
Free Online Tools — Convert, Compress, Transliterate & Transform
PixelsSuite is a fast, privacy-focused toolkit that runs entirely in your browser or on your device. It is built to handle everyday image, PDF, OCR, and text tasks with simple and reliable controls.
Resize images one by one or in bulk, compress JPG, PNG, and GIF files to reduce size while keeping good clarity, and convert images between JPG, PNG, and WebP. You can also crop images and adjust them using rotate and flip tools.
Manage documents by converting images to PDF, PDF to Word, and Word to PDF, or edit PDFs directly in your browser. Extract text from images using built-in OCR for quick and accurate results.
Convert Sinhala text between standard and chat formats, and use additional tools such as a meme generator, color picker, and other quick image utilities for simple edits.
Document Converter
Convert Image to PDF, PDF to Word, and Word to PDF quickly and securely.
Image → PDF
PDF → Word
Word → PDF
PDF Editor
Edit PDF files directly in your browser with simple tools and controls.
Open Editor →
Resize
Crop

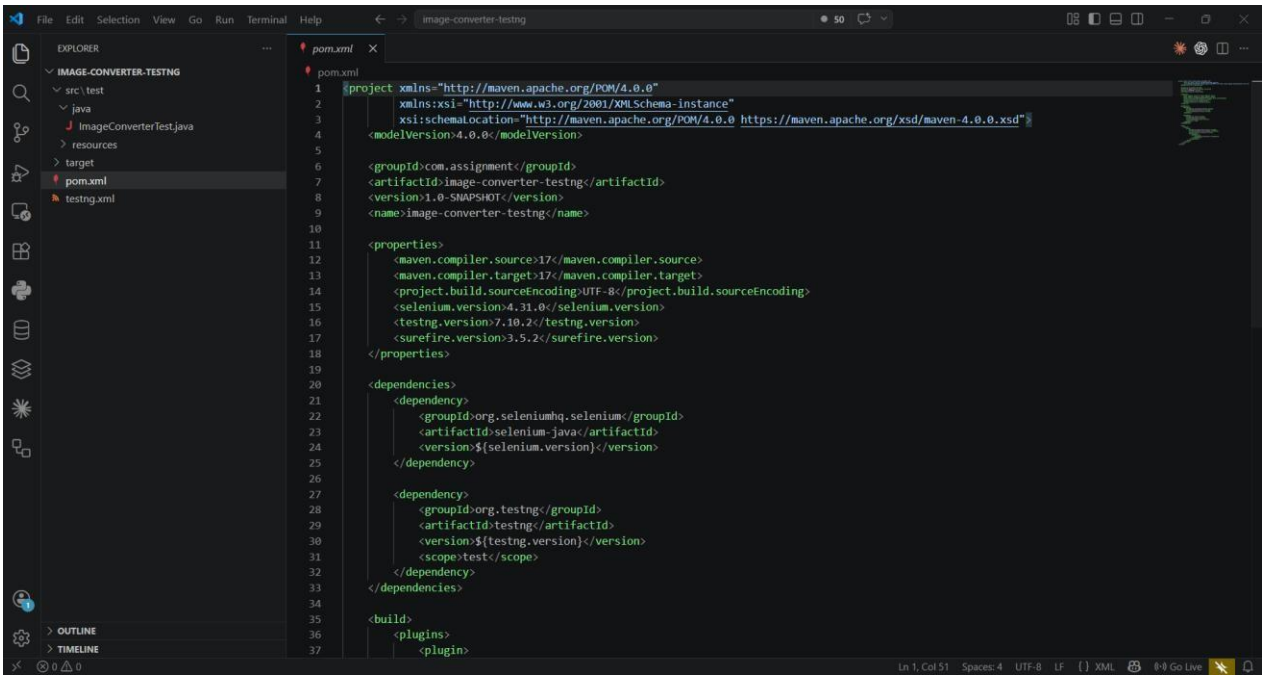
5.2 Test Execution

The automated test cases were executed using the **TestNG test runner** in a Chrome browser environment.

The execution was performed to validate the main functionality of the Image Converter module under normal usage conditions.

The following automated test cases were executed:

- **A/T-01** – Verify Image Converter page opens successfully
- **A/T-02** – Verify valid JPG upload works correctly



```java
import java.nio.file.Files;
import java.nio.file.Path;
import java.time.Duration;

import org.openqa.selenium.By;
import org.openqa.selenium.JavascriptExecutor;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import org.testng.Assert;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

public class ImageConverterTest {

    private WebDriver driver;
    private WebDriverWait wait;

    @BeforeMethod
    public void setup() {
        ChromeOptions options = new ChromeOptions();
        options.addArguments("--remote-allow-origins=*");

        driver = new ChromeDriver(options);
        driver.manage().window().maximize();

        wait = new WebDriverWait(driver, Duration.ofSeconds(20));
        driver.get("https://www.pixelssuite.com");
    }

    @AfterMethod
    public void teardown() {
        if (driver != null) {
```

```
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Apr 19, 2026 7:38:32 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
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Apr 19, 2026 7:38:32 PM org.openqa.selenium.chromium.ChromiumDriver lambda$new$4
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[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 20.43 s -- in TestSuite
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  22.771 s
[INFO] Finished at: 2026-04-19T19:38:51+05:30
[INFO] ------------------------------------------------------------------------

C:\Users\chamo\image-converter-testng>
```

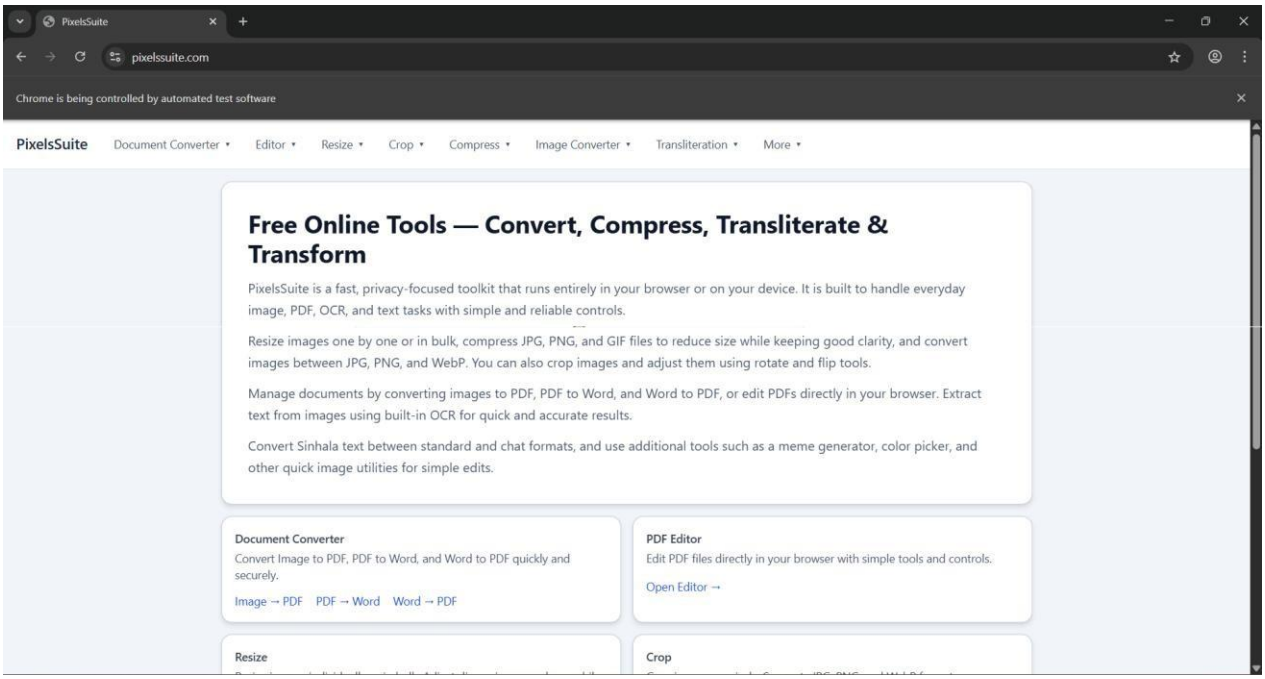

5.3 Test Report

The execution results showed that both automated test cases were completed successfully.
The generated output confirmed that the selected Image Converter functionality worked as expected for the tested scenarios.

5.4 Automated Test Cases

Automated Test ID	Test Scenario	Expected Result	Status
A/T-01	Verify Image Converter page opens successfully	Page loads and main sections are visible	Pass
A/T-02	Verify valid JPG upload works correctly	JPG file uploads successfully and the Download button becomes visible	Pass

6 Defects Found

Defect ID	Module	Description	Steps to Reproduce	Expected Behavior	Actual Behavior	Severity	Status
D-001	Image Converter	Clear button does not fully reset the page	Upload a valid JPG image and click Clear	The page should return to the initial empty state and remove the uploaded image from both Convert and Preview sections	The Convert section was reset, but the Preview section still displayed the previously uploaded image	Medium	Open